

Demonstrating coherent cross-chip quantum dynamics: an out-of-time-order-correlator experiment on a modular superconducting processor with QST interconnects

Draft proposal — cross_chips_sim

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Abstract

We propose a Loschmidt-echo out-of-time-order-correlator (OTOC) experiment on a modular processor consisting of four (extensible to five or six) 4-qubit chips connected by microwave cables. The key architectural fact is that the cables support two operating modes: a direct cross-chip R_{ZX} entangling gate with depolarizing error $\varepsilon_{RZX} \approx 0.1$, and full quantum state transfer (QST) with $\varepsilon_{QST} \approx 0.01$. We show that compiling *all* cross-chip interactions as QST round trips through dedicated port qubits reduces the effective cross-chip bond error from 0.1 to ≈ 0.025 , which is the difference between an operator light cone that fades below the shot-noise floor mid-module and one that coherently traverses all three cables. The echo protocol reads out a single qubit, making the modest readout fidelities (P_1 success 0.85–0.90) a trivially correctable scale factor, and its butterfly-free reference run doubles as a whole-machine benchmark and noise normalizer. We give the full compilation, qubit layout, error budget, anticipated (mock) figures, and extensions to 5–6 chips including a periodic ring and a “butterfly versus cable” race that directly visualizes the interconnect beating the emergent Lieb–Robinson velocity of the simulated dynamics.

1 Goal and headline claim

The purpose of this experiment is *not* to measure chemistry observables (as in the HF/VQE line of work) but to demonstrate that the machine performs **coherent many-body time evolution across chip boundaries**. The headline claim we want a referee to take away:

Quantum information injected on chip 1 spreads ballistically through Trotterized many-body dynamics, crosses three (or five) optical-quality cable links, and arrives on the last chip with measurable OTOC signal — and this is only possible because cross-chip gates are compiled through state transfer rather than the native noisy cross-chip entangler.

Two figures carry the paper: the light-cone heatmap $C(x, t)$ with the cable positions marked (Fig. 2), and the echo-fidelity comparison of the two compilation strategies (Fig. 3).

The document is self-contained: Sec. 2 defines every concept from scratch (Heisenberg operators, the OTOC, wave fronts, butterfly velocity, Lieb–Robinson bounds, scrambling, the Loschmidt echo); Sec. 3 works out an exact two-qubit toy model and a pen-and-paper argument for how the operator front propagates; the remaining sections specify the hardware, the compilation, and the experiment we actually propose to run.

2 Background: every object defined

All definitions introduced below are collected in the glossary, Table 1, for quick reference.

2.1 Heisenberg picture and operator spreading

Let $U(t) = e^{-iHt}$ (or a Trotterized circuit approximating it). A local operator W acting on a single site b evolves in the Heisenberg picture as

$$W(t) = U^\dagger(t) W U(t) = W + it [H, W] + \frac{(it)^2}{2!} [H, [H, W]] + \dots \quad (1)$$

For a Hamiltonian with only nearest-neighbor terms, each nested commutator in Eq. (1) can extend the *support* of the operator (the set of sites on which it acts nontrivially) by at most one lattice site. So $W(t)$ starts as a single-site operator and grows into an increasingly nonlocal one — this growth of support is called **operator spreading**, and it is the microscopic mechanism behind the quantum butterfly effect. Note that simple expectation values $\langle W(t) \rangle$ are blind to this growth; one needs a second operator to *probe* how far $W(t)$ has spread.

2.2 The OTOC, precisely

Take two Hermitian *and* unitary local operators (Paulis qualify): W on site b (the **butterfly operator**) and V on a distant site a (the **probe operator**). At $t = 0$ they act on disjoint sites, so $[W, V] = 0$ exactly. The quantity of interest is the **squared commutator**

$$C(t) = \frac{1}{2} \langle \psi | [W(t), V]^\dagger [W(t), V] | \psi \rangle \geq 0, \quad (2)$$

which measures, in the most literal operational sense, how much a perturbation applied at site b at time 0 disturbs a measurement at site a at time t . Expanding the commutator and using $W^2 = V^2 = \mathbb{I}$,

$$\boxed{C(t) = 1 - \text{Re } F(t), \quad F(t) = \langle \psi | W(t) V W(t) V | \psi \rangle.} \quad (3)$$

$F(t)$ is the **out-of-time-order correlator** — exactly the “ $1 - M(t) O M(t) O$ ” structure, with $M(t) = W(t)$ the evolved butterfly and $O = V$ the probe. The name comes from reading the time arguments of $W(t) V(0) W(t) V(0)$ left to right: $t \rightarrow 0 \rightarrow t \rightarrow 0$. This ordering does not appear in any ordinary (causal) response function, and it cannot be sampled by any single forward-in-time run — measuring $F(t)$ requires either an interferometric ancilla or *backward evolution* $U^\dagger$, which is why our protocol contains an echo.

Interpretation of the two limits:

- $F = 1, C = 0$: $W(t)$ still commutes with V ; the perturbation at b has not yet reached a . Information is still outside the light cone.
- $F \rightarrow 0, C \rightarrow 1$: in a chaotic system at late times $W(t)$ becomes an essentially random nonlocal operator, which on average half-commutes and half-anticommutes with V ; the OTOC decays to ≈ 0 and the squared commutator saturates. (Perfect anticommutation would give $F = -1, C = 2$; integrable dynamics can transiently reach such values and revive.)

2.3 Light cone, wave front, and butterfly velocity

Fix the butterfly site b and scan the probe site a ; write $x = |a - b|$ for their separation and regard $C(x, t)$ as a function on the space–time plane. In every locally interacting system it has the same shape:

- an **outside-the-cone** region ($x \gg v_B t$) where $C \approx 0$;
- a **wave front**: the narrow strip around $x \approx v_B t$ where C rises from ≈ 0 to its saturation value. The *front position* $x_f(t)$ (e.g. the contour $C = 0.5 C_{\text{sat}}$) moves ballistically, $x_f \simeq v_B t$, and the *front width* $w(t)$ is the distance over which the rise happens;
- a **saturated interior** ($x \ll v_B t$) where the operator has fully arrived.

The slope of the front defines the **butterfly velocity** v_B : the speed at which the influence of a local perturbation spreads. Near the front of a chaotic system the growth is often modeled as exponential, $C(x, t) \sim \epsilon e^{\lambda_L(t-x/v_B)}$, with λ_L a quantum analogue of a Lyapunov exponent (a form strictly derived in semiclassical/large- N settings; in spin chains the front is sharper and broadens only sub-ballistically); behind the front it saturates. A convenient phenomenological form, used for the mock figures below, is

$$C(x, t) \approx \frac{1}{2} C_{\text{sat}} \left[1 - \tanh\left(\frac{x - v_B t}{w}\right) \right]. \quad (4)$$

In a brickwork Trotter circuit, time is measured in steps S and v_B in sites per step; for the kicked Ising model at generic parameters $v_B \approx 1$ site/step.

2.4 Lieb–Robinson bound

The Lieb–Robinson theorem states that for any Hamiltonian with bounded local interactions there exist constants c, ξ, v_{LR} such that

$$\| [W_b(t), V_a] \| \leq c \|W\| \|V\| e^{-(|a-b| - v_{LR} t)/\xi}. \quad (5)$$

In words: even though the Schrödinger equation is not relativistic, locally interacting quantum matter has an *emergent* speed limit v_{LR} ; outside the cone $|a - b| > v_{LR} t$ all commutators — and hence the OTOC signal, Eq. (2) — are exponentially small. The butterfly velocity always satisfies $v_B \leq v_{LR}$. For circuits the statement is even sharper and needs no theorem: after S brickwork layers the support of $W(S)$ has grown by *at most* S sites (a strict, not just exponential, light cone). Two remarks that matter for this proposal:

1. The measured OTOC front (Fig. 2) is a direct experimental image of an emergent Lieb–Robinson cone.
2. The bound constrains the *simulated* dynamics only. Hardware channels that are not part of the Hamiltonian — our QST cables — are free to move quantum information faster than v_{LR} of the simulated model. That contrast is the point of the “butterfly–cable race” (Sec. 9.3).

2.5 Scrambling, and why it is not decoherence

Scrambling is the process by which initially local information becomes encoded in many-body correlations spread over the entire system: no local measurement can recover it, but it is still there — the global evolution is unitary. The OTOC saturating ($F \rightarrow 0$) behind the front is the standard operational diagnostic of scrambling. Decoherence, by contrast, leaks information into the environment irreversibly. The echo protocol distinguishes the two beautifully: backward evolution $U^\dagger$ *refocuses* scrambled information (it is unitary and we control it) but cannot refocus decohered information. This is precisely why the butterfly-free echo run measures hardware fidelity, while the ratio of the two runs isolates the scrambling physics.

2.6 Loschmidt echo, and the exact identity behind our measurement

A **Loschmidt echo** is any forward-then-backward protocol $U^\dagger P U$ with a perturbation P inserted at the turning point. Our measurement is a single-qubit observable at the end of such an echo, and it equals the OTOC *exactly* — no approximation. Derivation: prepare $|\psi_0\rangle = |0 \cdots 0\rangle$, evolve forward, apply the butterfly $W = X_b$, evolve backward, measure Z_a . The final state is $|\phi\rangle = U^\dagger X_b U |\psi_0\rangle$, so the measured expectation is

$$\langle Z_a \rangle_{\text{echo}} = \langle \psi_0 | (U^\dagger X_b U)^\dagger Z_a (U^\dagger X_b U) | \psi_0 \rangle = \langle \psi_0 | W(t) Z_a W(t) | \psi_0 \rangle, \quad (6)$$

with $W(t) = U^\dagger X_b U$ the Heisenberg butterfly. Now use $Z_a |\psi_0\rangle = + |\psi_0\rangle$ (the initial state is a Z_a eigenstate) to append a free Z_a on the right:

$$\boxed{\langle Z_a \rangle_{\text{echo}} = \langle \psi_0 | W(t) Z_a W(t) Z_a | \psi_0 \rangle = F(t) \implies C(t) = 1 - \langle Z_a \rangle_{\text{echo}}.} \quad (7)$$

So the single number we read out *is* the out-of-time-order correlator $F(t) = \langle W(t) V W(t) V \rangle$ with $W = X_b$, $V = Z_a$ — the “ $1 - M(t)OM(t)O$ ” form — and the butterfly-free reference run (which would ideally give $\langle Z_a \rangle = 1$ since $U^\dagger U = \mathbb{I}$) measures the hardware echo fidelity used for normalization. The identity holds for any Pauli probe V , provided the initial state is prepared as a +1 eigenstate of V (e.g. prepare $|+\rangle$ on site a and measure X_a if one wants $V = X_a$); we use $V = Z_a$ and $|0 \cdots 0\rangle$ throughout. The identity holds verbatim for any Pauli probe V , provided the initial state is a +1 eigenstate of V (e.g. prepare $|+\rangle$ on site a and measure X_a); this generalization is used in the toy-model check below.

2.7 Trotterization

We cannot apply e^{-iHt} directly; we approximate it by S discrete steps, $U(t) \approx (U_{\text{step}})^S$ with $\delta t = t/S$, where U_{step} applies the ZZ bond layer and the single-qubit field layer in sequence. The per-step error is $O(\delta t^2 \|[H_{ZZ}, H_X]\|)$; for OTOC physics this is benign because a Trotter circuit is itself a perfectly valid locally interacting dynamics (“kicked Ising”) with its own front, v_B , and scrambling — we study the circuit as the model, not as an approximation to be apologized for.

Term	Meaning
Operator spreading	Growth of the support of a Heisenberg operator $W(t)$, Eq. (1).
Butterfly operator W	The local perturbation (X_b here) whose influence we track.
Probe operator V	The distant local observable (Z_a here) used to detect arrival.
OTOC $F(t)$	$\langle W(t) V W(t) V \rangle$; time order $t \rightarrow 0 \rightarrow t \rightarrow 0$; requires backward evolution.
Squared commutator $C(t)$	$\frac{1}{2} \langle [W(t), V] ^2 \rangle = 1 - \text{Re} F$; the “how much does b affect a ” number.
Light cone	Space–time region $x \lesssim v_B t$ where C is appreciably nonzero.
Wave front	The moving strip where C rises from 0 to saturation; position $x_f \simeq v_B t$, width w .
Butterfly velocity v_B	Speed of the front; $v_B \approx 1$ site/step for kicked Ising.
Lieb–Robinson bound	Emergent speed limit $v_{LR} \geq v_B$ for any local dynamics, Eq. (5).
Scrambling	Local info hidden in global correlations; diagnosed by $F \rightarrow 0$ behind the front.
Loschmidt echo	Forward–perturb–backward protocol; its fidelity isolates hardware noise.
Trotter step	One ZZ -layer + field-layer application; the discrete time unit of the experiment.

Table 1: Quick-reference glossary for all terms used in this proposal.

2.8 Glossary

3 Toy model: the OTOC by hand

3.1 Two qubits, exactly solvable

Take $H = J Z_1 Z_2$, butterfly $W = X_1$, probe $V = X_2$, state $|\psi_0\rangle = |00\rangle$. Because X_1 anticommutes with $Z_1 Z_2$,

$$W(t) = e^{iHt} X_1 e^{-iHt} = X_1 e^{-2iJt Z_1 Z_2} = \cos(2Jt) X_1 - \sin(2Jt) Y_1 Z_2. \quad (8)$$

Read Eq. (8) as operator spreading in miniature: at $t = 0$ the operator lives on site 1 only; under evolution it rotates weight into $Y_1 Z_2$, a two-site operator. The commutator with the probe is carried entirely by that grown component:

$$[W(t), X_2] = -\sin(2Jt) [Y_1 Z_2, X_2] = -2i \sin(2Jt) Y_1 Y_2, \quad (9)$$

so the squared commutator and OTOC are

$$C(t) = \frac{1}{2} \cdot 4 \sin^2(2Jt) \langle (Y_1 Y_2)^2 \rangle = 2 \sin^2(2Jt) = 1 - \cos(4Jt), \quad \text{Re } F(t) = \cos(4Jt). \quad (10)$$

Sanity checks: $C(0) = 0$ (disjoint operators commute); C first becomes appreciable on the coupling timescale $1/J$; at $2Jt = \pi/2$ we get $F = -1$, $C = 2$ — maximal anticommutation, the strongest possible butterfly effect; and the dynamics is integrable (one bond!), so C revives periodically instead of saturating. Note the probe choice matters: $V = Z_2$ would give $C \equiv 0$ here, because the grown component $Y_1 Z_2$ commutes with Z_2 — for this Hamiltonian all spreading happens in the X/Y sector of site 2. Running the echo protocol of Sec. 2.6 with the matching probe — prepare $|0\rangle \otimes |+\rangle$, evolve, kick X_1 , unevolve, measure X_2 — reproduces Eq. (10) exactly (verified numerically), and is a recommended first simulation/hardware check. In the full kicked-Ising experiment the R_x and R_z kicks mix all Pauli sectors, so the standard choice $V = Z_a$, $|\psi_0\rangle = |0 \cdots 0\rangle$ detects the front at generic parameters.

3.2 Why the front moves one site per step

Now chain the toy: $H_{ZZ} = J \sum_i Z_i Z_{i+1}$ alone does not scramble. Equation (8) generalizes to $X_1(t) = \cos X_1 - \sin Y_1 Z_2$, and there it stops: growing past site 2 would require a nonzero commutator with the next bond $Z_2 Z_3$, but both X_1 and $Y_1 Z_2$ commute with it (the only piece touching site 2 is Z_2 , which commutes with $Z_2 Z_3$), so the operator just rotates inside $\text{span}\{X_1, Y_1 Z_2\}$ forever and the support never passes site 2. The transverse kick is what keeps the front moving: a field layer $e^{-i\theta X_2/2}$ rotates $Z_2 \rightarrow \cos \theta Z_2 + \sin \theta Y_2$, and the Y_2 component *anticommutes* with $Z_2 Z_3$, so the next ZZ layer grows the support to site 3. By induction, each (bond layer + kick layer) step advances the support edge by at most one site — a two-line proof of the strict circuit light cone of Sec. 2.4 — and at generic angles it advances by essentially exactly one site, giving $v_B \approx 1$ site/step. The front is not perfectly sharp because the *weight* carried by the newest edge operators is a product of sin/cos factors accumulated along the way; summing the leading paths gives a front of finite width w that broadens sub-ballistically, which is what Eq. (4) parameterizes and Fig. 2 displays.

3.3 From toy to proposal

The proposed experiment is this toy model scaled up: the same ZZ bonds (three of which happen to cross chip boundaries and are compiled through QST, Sec. 5), the same kicks, the same echo measurement of Eq. (7). Everything measurable reduces to the objects defined above: the heatmap of $C(a, S)$ shows the light cone, its slope is v_B , its rise is the wave front, its plateau is scrambling, and the butterfly-free denominator is the Loschmidt-echo fidelity of the machine.

4 Hardware model

4.1 Connectivity

Each chip is a 2×2 plaquette; chips are chained by single cables attached to one “port-adjacent” qubit per side (Fig. 1). Cable links (4-chip machine): (4, 5), (7, 9), (11, 13).

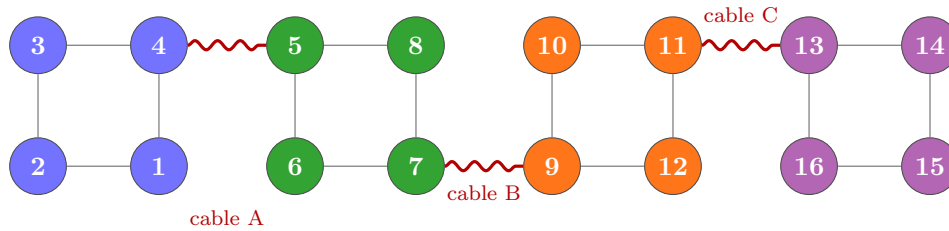


Figure 1: Four-chip machine. Wavy red edges are cables supporting either a noisy R_{ZX} ($\varepsilon_{RZX} \approx 0.1$) or clean QST ($\varepsilon_{QST} \approx 0.01$).

Channel	Symbol	Value
Intra-chip 2q depolarizing	ε_{2q}	0.005
Intra-chip 1q depolarizing	ε_{1q}	10^{-4}
Cross-chip R_{ZX} depolarizing	ε_{RZX}	0.1
QST (one transfer over cable)	ε_{QST}	0.01
Readout success $P(\text{read } 0 0)$	P_0	0.92–0.97 (per qubit)
Readout success $P(\text{read } 1 1)$	P_1	0.85–0.90 (per qubit)

Table 2: Error parameters. QST additionally requires the cable mode and the receiving qubit to start in $|0\rangle$ (single-excitation constraint of the bus).

4.2 Noise model (numbers taken from the simulation stack)

5 Compiling cross-chip interactions

5.1 The QST routing gadget (the workhorse)

To apply $ZZ(\theta) = e^{-i\theta Z \otimes Z/2}$ between data qubit a (chip k , cable endpoint) and data qubit b (chip $k+1$, neighbor of the port p which is held in $|0\rangle$):

$$\boxed{\text{QST}_{a \rightarrow p} \longrightarrow ZZ(\theta)_{p,b} \text{ (local)} \longrightarrow \text{QST}_{p \rightarrow a}} \quad (11)$$

The return transfer automatically re-empties p , so the gadget is self-restoring and needs *no* measurement, reset, or feed-forward. Effective bond error:

$$\varepsilon_{\text{bond}} \approx 2\varepsilon_{QST} + \varepsilon_{2q} \approx 0.025 \quad \text{vs.} \quad \varepsilon_{RZX} \approx 0.1. \quad (12)$$

A $4\times$ error reduction per cross-chip bond, which compounds exponentially with circuit depth.

5.2 Rejected alternatives (and why)

- **Direct cross-chip R_{ZX} :** $\varepsilon_{RZX} = 0.1$ per bond. With three cables and an echo circuit of depth $2S$, the cross-chip error budget alone is $0.6S$; the echo falls to the shot-noise floor by $S \approx 7$, before the light cone can cross the last cable (Sec. 7).
- **Teleported gates via QST/2 Bell pairs:** gate teleportation consumes a Bell pair *plus a mid-circuit measurement and feed-forward*. With P_1 success of 0.85–0.90, each teleported gate inherits a 10–15% Pauli error from readout — *worse than the raw R_{ZX}* . Bell-pair gates become attractive only if readout exceeds $\sim 99\%$.
- **Randomized-measurement OTOCs** (no backward evolution needed): require correlating full bitstrings over many qubits, so the $\sim 10\%$ per-qubit readout error compounds catastrophically. Not viable here.

6 The experiment: echo OTOC of a kicked Ising chain

6.1 Model and Trotterization

We simulate a kicked/transverse-field Ising chain on N data qubits,

$$H = J \sum_{\langle i,j \rangle} Z_i Z_j + h_x \sum_i X_i + h_z \sum_i Z_i, \quad (13)$$

with one first-order Trotter step $U_{\text{step}} = [\prod_{\langle ij \rangle} ZZ(2J\delta t)] [\prod_i R_x(2h_x\delta t)R_z(2h_z\delta t)]$. This is exactly the chained toy model of Sec. 3: the ZZ layer grows operators across bonds, the kick layer keeps the front moving. The point $h_z = 0$ is integrable (maps to free fermions; sharp ballistic front with revivals); $h_z \neq 0$ breaks integrability (chaotic; front broadens, OTOC saturates behind it — genuine scrambling). Cross-chip ZZ bonds use the gadget of Eq. (11); everything else is native intra-chip.

6.2 Echo (Loschmidt) OTOC protocol

We run the echo protocol of Sec. 2.6, which by the exact identity Eq. (7) measures the OTOC of Sec. 2.2, $F = \langle W(t) V W(t) V \rangle$, with butterfly $W = X_b$ and probe $V = Z_a$:

1. Prepare $|\psi_0\rangle = |0 \cdots 0\rangle$ on the data chain (a Z_a eigenstate for every a , as the identity requires).
2. Evolve forward: $U = (U_{\text{step}})^S$.
3. Apply the butterfly X_b on the chip-1 end of the chain (or skip it, for the reference run).
4. Evolve backward: $U^\dagger$ (reversed circuit, negated angles — the QST gadget reverses trivially).
5. Measure $\langle Z_a \rangle$ on a *single* qubit a ; sweep a over the chain and S over Trotter depths.

Noiselessly, $\langle Z_a \rangle_{\text{with } X_b} = F(a, S)$ and $\langle Z_a \rangle_{\text{without } X_b} = 1$. With hardware noise, both runs acquire (to first order in depolarizing noise) the same fidelity prefactor f_{echo} , so we report the ratio-normalized squared commutator

$$C(a, S) = 1 - \frac{\langle Z_a \rangle_{\text{with } X_b}}{\langle Z_a \rangle_{\text{without } X_b}} \approx 1 - \frac{f_{\text{echo}} F(a, S)}{f_{\text{echo}}} = 1 - F(a, S), \quad (14)$$

i.e. precisely the $1 - \langle W(t) V W(t) V \rangle$ of Eq. (3). The denominator is simultaneously (i) the whole-circuit Loschmidt-echo fidelity — a free whole-machine benchmark — and (ii) the normalizer that cancels the depolarizing background, the standard trick that makes OTOCs unusually noise-robust. Because only one qubit is ever read out, readout error reduces to a known 2×2 confusion matrix (from P_0, P_1 of Table 2) and is inverted exactly.

6.3 Qubit layout (4-chip machine)

Role	Chip 1	Chips 2–3	Chip 4
Data chain (11 qubits)	3–2–1–4	6–7 and 12–11	16–15–14
Ports (held in $ 0\rangle$)	—	5 and 9	13
Spares	—	8 and 10	—

Table 3: Chain order: $3, 2, 1, 4 \rightsquigarrow 6, 7 \rightsquigarrow 12, 11 \rightsquigarrow 16, 15, 14$. Virtual bonds: $4 \leftrightarrow 6$ via port 5; $7 \leftrightarrow 12$ via port 9; $11 \leftrightarrow 16$ via port 13. Butterfly qubit $b = 3$ (chain position 1).

7 Error budget and feasibility

The 11-site open chain of Table 3 has 10 bonds: 7 intra-chip (3–2, 2–1, 1–4; 6–7; 12–11; 16–15, 15–14) and 3 virtual. Each step also applies two field rotations (R_x , R_z) per qubit. Per Trotter step:

$$\varepsilon_{\text{step}}^{\text{QST}} \approx 3 \times 0.025 + 7 \times 0.005 + 2 \times 11 \times 10^{-4} \approx 0.11, \quad (15)$$

$$\varepsilon_{\text{step}}^{\text{RZX}} \approx 3 \times 0.10 + 7 \times 0.005 + 2 \times 11 \times 10^{-4} \approx 0.34. \quad (16)$$

Echo circuits have depth $2S$, so the expected echo fidelity is $F(S) \approx e^{-2S\varepsilon_{\text{step}}}$. With $v_B \approx 1$ site/step the front sits near chain position $1 + S$ (cables A, B, C lie after positions 4, 6, 8):

S	F_{QST}	F_{RZX}	Light-cone reach (butterfly at position 1)
3	0.51	0.13	position 4: crossing cable A into chip 2
5	0.33	0.034	position 6: crossing cable B into chip 3
8	0.17	0.0045	position 9: crossing cable C into chip 4
10	0.11	0.0012	position 11: end of chain

With 8192 shots the statistical floor on $\langle Z \rangle$ is $\approx 1/\sqrt{8192} \approx 0.011$. The QST compilation stays an order of magnitude above the floor out to $S = 10$; the R_{ZX} compilation reaches the floor near $S \approx 7$ and is already marginal ($3\times$ the floor) by $S = 5$ — before the butterfly crosses cable C. **This contrast is the experiment.**

8 Anticipated results (mock figures)

All plots below are drawn from analytic mock data — the tanh front of Eq. (4) with $v_B \approx 1$ site/step and front width $w \approx 1.4$ sites — purely to fix the visual target.

8.1 Money plot 1: the cross-chip light cone

8.2 Money plot 2: compilation strategy comparison

8.3 Supporting plot: chaotic vs. integrable spreading

9 Expanding to 5–6 chips: the fun part

With 5–6 chips the chain extends to ~ 13 – 15 data qubits (interior chips contribute 2 data + 1 port + 1 spare each), and several qualitatively new experiments open up.

9.1 Longer light cone and scrambling saturation

Five cables instead of three. The error budget scales gently: each extra interior chip adds two data sites, i.e. one virtual bond ($+0.025/\text{step}$), one local bond ($+0.005/\text{step}$), and two qubits of field-layer noise. At 6 chips (15-site chain: 9 intra-chip + 5 virtual bonds) $\varepsilon_{\text{step}} \approx 5 \times 0.025 + 9 \times 0.005 + 2 \times 15 \times 10^{-4} \approx 0.17$, so the echo at $S = 10$ is $e^{-3.5} \approx 3\%$ — marginal but $\sim 5\times$ above the floor with 3×10^4 shots ($1/\sqrt{3 \times 10^4} \approx 0.006$). The payoff: enough depth *and* length to see the OTOC front pass *and* saturate behind it, i.e. genuine scrambling, not just transport.

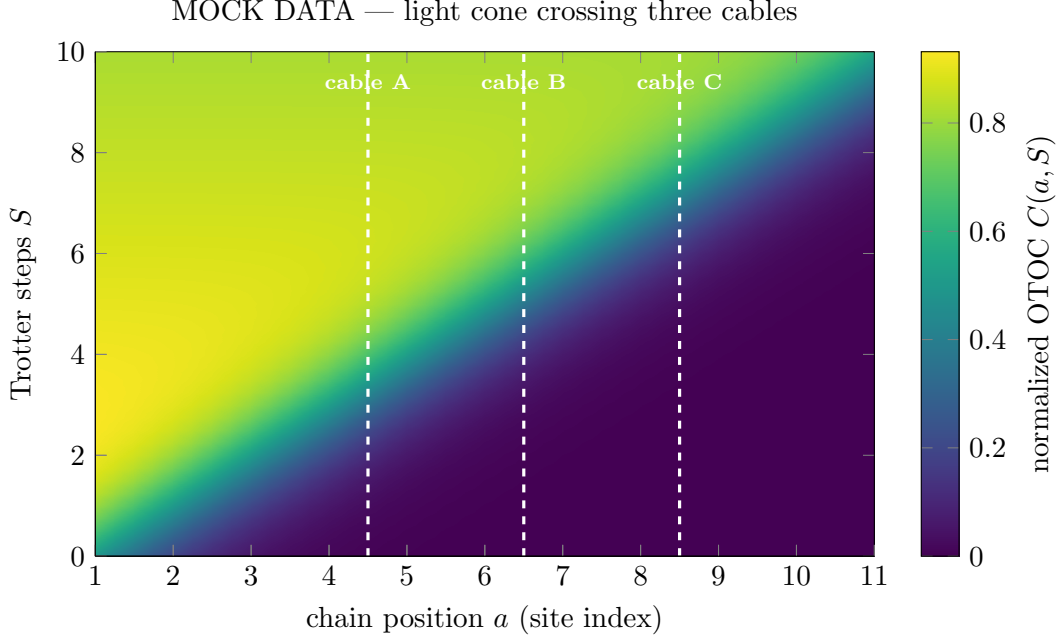


Figure 2: [MOCK] Normalized OTOC $C(a, S)$, butterfly on site 1 (qubit 3); light cone, wave front, and v_B as defined in Sec. 2.3. The ballistic front crosses all three cable links (dashed) without visible degradation — the direct visual proof of coherent cross-chip dynamics. The same plot compiled with direct R_{ZX} would fade to the noise floor around cable B (Sec. 7).

9.2 Ring closure: periodic boundary conditions

If chip 6 can be cabled back to chip 1, the machine realizes a *periodic* chain — something planar fixed layouts cannot do without massive swap overhead. Signatures: the two counter-propagating OTOC fronts launched by one butterfly *collide on the far side of the ring*, giving a distinctive interference bump at the antipodal site at $t \approx L/(2v_B)$; momentum quantization becomes visible in quench spectroscopy. This is a demo only a modular machine does cheaply.

9.3 The butterfly–cable race (recommended headline demo)

Run two protocols simultaneously on one machine:

- **Lane 1 (dynamics):** the butterfly spreads through Trotter evolution at the emergent Lieb–Robinson velocity $v_B \approx 1$ site/step.
- **Lane 2 (interconnect):** a reference excitation is relayed down the *spare/port qubits* by bare QST hops, at one *chip* per hop.

9.4 Colliding butterflies

With ≥ 5 chips there is room to launch butterflies from *both* ends and watch the two OTOC fronts collide mid-machine (on a middle chip). The collision region shows enhanced operator weight and, at the integrable point, transmission of fronts through each other — a clean, visual many-body scattering experiment.

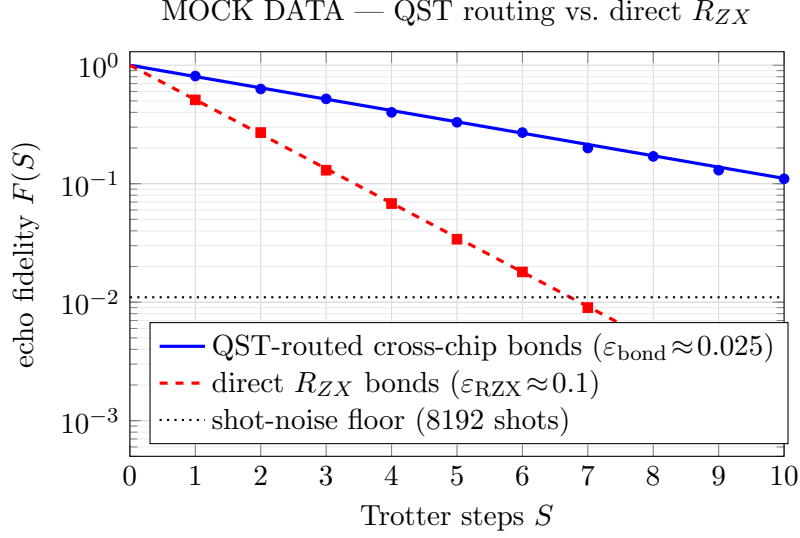


Figure 3: [MOCK] Butterfly-free echo fidelity vs. depth for the two compilations of the *same* physical circuit. The gap is the quantitative “power of the interconnect” statement.

10 Error mitigation stack

1. **Readout:** exact 2×2 confusion-matrix inversion on the single measured qubit, calibrated from P_0, P_1 .
2. **Normalization:** Eq. (14) cancels the depolarizing background to first order.
3. **ZNE on the cables:** inserting identity QST round-trip pairs ($a \rightarrow p \rightarrow a$) scales cross-chip noise in clean units of $2\varepsilon_{\text{QST}}$ — an unusually well-controlled noise-amplification knob unique to this architecture.
4. **CDR:** the existing Clifford-data-regression machinery (`main_cursor_lib`) applies directly since the Trotter circuits are Clifford at $J\delta t, h\delta t \in \{0, \pi/4\}$.

11 Milestones

1. **M1 — gadget validation:** single virtual bond (chips 1–2); process fidelity of Eq. (11) vs. direct R_{ZX} .
2. **M2 — domain-wall quench:** prepare $|0 \cdots 0 1 \cdots 1\rangle$, forward evolution only, measure $\langle Z_i(t) \rangle$ profile crossing the cables. Half the depth of the echo; validates layout, calibration, and mitigation.
3. **M3 — echo OTOC, 4 chips:** Figs. 2 and 3; then the integrable-vs-chaotic sweep of Fig. 4.
4. **M4 — 5/6 chips:** extended light cone; ring and/or butterfly–cable race (Fig. 5); colliding butterflies.

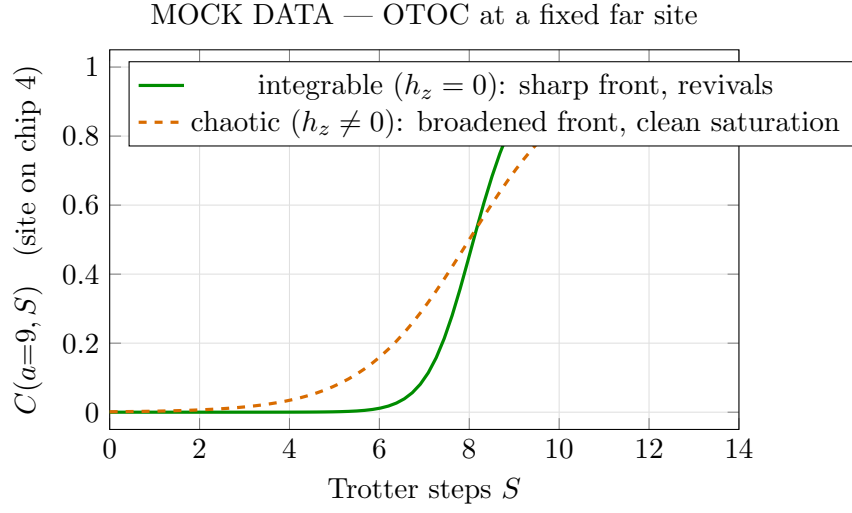


Figure 4: [MOCK] Fixing the measurement qubit on chip 4 and sweeping depth distinguishes integrable and chaotic dynamics — physics content beyond a hardware benchmark.

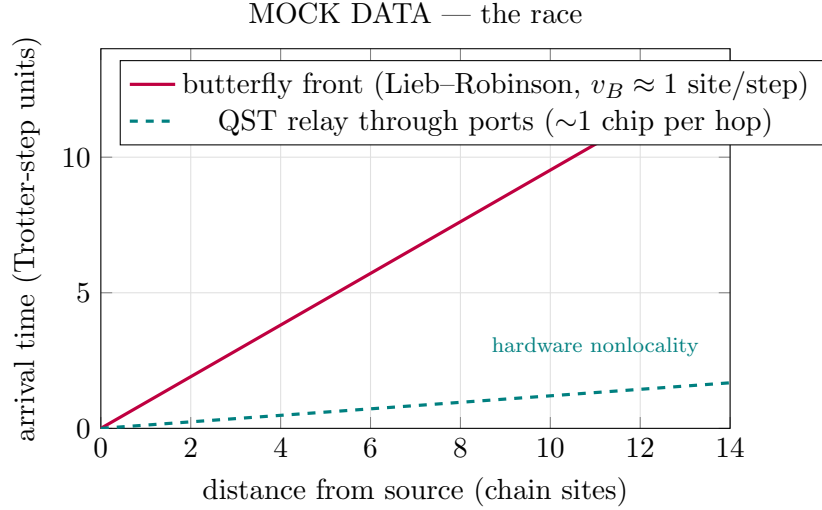


Figure 5: [MOCK] Arrival time vs. distance for the two lanes. The interconnect beats the light cone of the simulated Hamiltonian by an order of magnitude — a one-plot statement that modular hardware adds a resource (long-range quantum channels) that the simulated physics does not have.